ATLAS — Aerospace Supply Chain Intelligence

Executive Product Brief

Platform: Multi-Agent AI Copilot for Aerospace Supply Chain Operations
AI Assistant: Atlas AI Orchestrator
Organization: Heroux-Devtek Inc.
Framework: Microsoft Agent Framework & Microsoft Foundry
AI Models: GPT-5.4-mini, KIMI2.6 (via Microsoft Foundry)
Version: 1.1 — Proof of Concept
Date: June 2026

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1. Executive Summary

Heroux-Devtek is the world's third-largest aerospace landing gear manufacturer, operating 8 facilities across Canada, USA, UK, and Spain — serving programs including 777X, A350, A220, 737MAX, F-35, CH-53K, and Global 7500. The current S&OP environment relies on disconnected systems for demand planning, inventory management, supplier monitoring, and production scheduling — resulting in reactive decision-making, stockout events, and significant manual overhead.

The **ATLAS Supply Chain Intelligence Platform** introduces a multi-agent AI orchestration layer that transforms how S&OP planners, procurement managers, operations directors, and leadership interact with supply chain data. The platform uses 4 specialist AI agents with a 5-stage scenario pipeline to deliver quantitative what-if simulation, predictive inventory risk scoring, automated replenishment recommendations, contract compliance validation, labor utilization tracking, and multi-format report generation — all through a trilingual (English/French/Spanish) conversational interface.

> **Key Differentiator:** Every page in ATLAS is an Atlas AI-powered workflow. The AI doesn't decorate the app — it drives it. The entire UI adapts dynamically based on the user's language preference, with all labels, charts, tables, and AI responses rendered in the selected language.

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2. The Problem

2.1 Current State Challenges

Domain	Challenge	Impact
Demand Planning	Forecasts in separate systems, manual reconciliation	Misaligned production, missed program rate changes
Inventory Management	Reactive stockout detection, no predictive visibility	CAD $2M+/month in expedite costs

Scenario Analysis	Manual spreadsheet modeling for what-if simulations	2-3 days per scenario, decisions without quantitative backing
Report Generation	Weekly S&OP decks assembled manually from multiple sources	6+ hours per deck, 40% of planner time on formatting
Supplier Monitoring	No real-time view of cascading supply risks	Titanium allocation cuts discovered when forging lines idle
Contract Compliance	PO pricing checked via monthly spot-audits	Margin erosion on long-term programs
Labor Tracking	Shift-level efficiency scattered across 8 facilities	Cannot identify underutilization or skill bottlenecks
Global Communication	Multi-site teams across Canada, US, UK, Spain	Language barriers delay decisions between sites

2.2 Core Pain Points

1. **Reactive planning** — stockouts detected after the fact, not 8 weeks in advance
2. **Data fragmentation** — S&OP meetings rely on stale Excel snapshots from 5+ systems
3. **Scenario paralysis** — what-if questions take days, so decisions are made on intuition
4. **Report drudgery** — senior planners spend 40% of time formatting, not analyzing
5. **Supplier blindness** — no cascading risk visibility across the 10-supplier network
6. **Contract leakage** — PO overages go undetected between quarterly reviews
7. **Language friction** — CESA Spain teams require translation for every English report

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3. Solution Architecture

3.1 Design Philosophy

The platform operates as an **AI-driven decision engine**:

- **AI drives the application, not decorates it** — every page is an Atlas AI-powered workflow
- **Microsoft Agent Framework & Microsoft Foundry** for multi-agent orchestration
- **4 specialist agents + orchestrator** running in parallel
- **5-stage scenario pipeline** for quantitative what-if simulation
- **Trilingual operation** — all UI, charts, tables, and AI responses adapt to EN/FR/ES
- **All agent output is real** (LLM-generated) — only underlying data is synthetic for this PoC
- **Conversational interface** as the primary interaction mode

3.2 Platform Architecture

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│              Frontend — React 19 + TypeScript + Vite + Fluent UI 9    │
│   10 Module Pages │ Atlas AI Chat Panel │ Agent Results │ Report Gen   │
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                                 │ SSE Streaming (/api/chat)
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│                    Backend — FastAPI + Agent Framework                      │
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│  ┌────────────────────────────────────────────────────────────────────┐  │
│  │   Atlas AI Orchestrator (Microsoft Agent Framework & Foundry)       │ │
│  │     Models: GPT-5.4-mini + KIMI2.6 (via Microsoft Foundry)              │
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│  │     Routes to specialist agents based on query classification      │ │
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│             │                                                              │
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│  │  Specialist Agents (4)                  Tools & Pipelines           │ │
│  │  • Demand Sensing Specialist            • Contract Price Validation │ │
│  │  • Inventory Risk Specialist            • PO Compliance Checker     │ │
│  │  • Supply Constraint Specialist         • Labor Utilization Dash    │ │
│  │  • Replenishment Specialist             • Scenario Pipeline (5-stg) │ │
│  │                                         • Report Gen (5 agents)     │ │
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│             │                                                              │
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│  │  Data Service Layer                     Document Generation          │ │
│  │  • SKU Master (50 assemblies)           • PPTX Pipeline (5 agents)  │ │
│  │  • Inventory Positions                  • DOCX/PDF Renderer         │ │
│  │  • Supplier Network (10)                • XLSX Workbooks            │ │
│  │  • Production Schedule (8 lines)        • HD Navy/Gold Branding     │ │
│  │  • Labor Records (8 facilities)                                     │ │
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4. Product Modules

4.1 KPI Dashboard

Capability	How It Works
Real-time KPI cards	8 metrics: MAPE, DOS, Fill Rate, OTD, Labor Util, Contract Compliance, Alerts, Production
Risk matrix	Pie chart: critical, warning, normal, excess inventory distribution
Critical alerts	Top active alerts with severity badges and affected programs
Context-aware chips	Page-specific quick actions adapt to current module and language

4.2 Demand Forecast

Capability	How It Works
8-week forecast	SKU-level point forecast with 80% and 95% confidence bands

Program change flags	Visual indicators when OEM rate changes affect demand
Demand signals	Trend direction, program schedules, MRO forecast signals
Translated legends	Chart legends and stat cards render in selected language

4.3 Inventory Health

Capability	How It Works
Risk scoring	Every SKU scored: critical (stockout imminent), warning, normal, excess
DOS-sorted view	Lowest days-of-supply first for immediate triage
Cert expiry tracking	Material certifications approaching expiry flagged for re-test
Badge translation	Risk levels display in active language (Critical/Critique/Critico)

4.4 Supply Network

Capability	How It Works
Supplier monitoring	10 aerospace suppliers: reliability scores, lead times, active orders
Production capacity	Horizontal bar chart of 8-line utilization with threshold coloring
Certification tracking	NADCAP, AS9100 approval status per supplier
Detail panels	Click-through for performance trends, PO history, quality metrics

4.5 Scenario Planner (Flagship)

Capability	How It Works
5-stage AI pipeline	Planner → Impact Analyzer → Mitigation Designer → Risk Assessor → Synthesizer
Quick scenario chips	Pre-built aerospace scenarios (rate increase, forging delay, AOG, titanium disruption)
Custom natural language	Describe any what-if in any supported language
4 result tabs	Impact Summary, Affected Part Numbers, Stock Timeline, Mitigation & Supply
Scenario comparison	Select any 2 historical scenarios for side-by-side KPI delta analysis
Full translation	Every tab, chart, table, badge renders in active language

4.6 Replenishment Plan

Capability	How It Works
Prioritized action cards	SKU, urgency badge, recommended qty, confidence, supplier, rationale
Scenario variants	Conservative, balanced, aggressive options with KPI trade-offs
Approve/dismiss workflow	One-click approval with KPI impact confirmation dialog

Translated dialogs	All button text, labels, and rationale in active language

4.7 Production Priorities

Capability	How It Works
Utilization chart	All 8 lines visualized with current utilization %
Schedule table	Line, plant, capacity, current SKU, shift pattern, maintenance
Capacity headroom	Spare capacity available for surge production identified

4.8 Labor Utilization

Capability	How It Works
Facility coverage	All 8 global facilities: Longueuil, Kitchener, Springfield, Nottingham, Laval, Livonia, Getafe, Seville
KPI cards	Average efficiency, direct labor %, total headcount, overtime hours
Facility comparison	Bar chart comparing all 8 sites
Filterable records	Daily records by facility, shift, skill category, overtime

4.9 Report Builder (Multi-Agent)

Capability	How It Works
7 templates	Weekly S&OP (PPTX), Inventory Status (PPTX), Executive Summary (DOCX/PDF), Replenishment (XLSX), Supplier Scorecard (XLSX)
5-agent pipeline	Planner → Content Writer → Designer → Critic → Repair (up to 2 iterations)
HD branding	Navy/gold corporate identity, logo, consistent formatting
Language-aware	Report content generated in selected language

4.10 Atlas AI Chat Panel

Capability	How It Works
Page-aware context	Welcome message and chips change based on active page
Trilingual interface	All UI elements adapt to EN/FR/ES
Agent visualization	Real-time planning blocks show active specialist and tool
Dynamic follow-ups	Post-response suggestion chips based on AI analysis
Message actions	Copy, upvote, downvote, regenerate on every response
Language-aware LLM	Responds in user's language; internal prompts stay English for optimal reasoning

5. Personas & Use Cases

The platform serves **5 core personas** across Heroux-Devtek's global operations:

Persona	Role	Key Journeys
S&OP Planner	Daily operations owner	Morning brief, demand analysis, scenario simulation, replenishment approval, report generation
VP Supply Chain	Executive decision-maker	Risk escalation, financial exposure assessment, board reporting, mitigation approval
Procurement Manager	Supplier & contract owner	Contract validation, PO compliance, emergency ordering, supplier scorecards
Operations Director	Production & labor	Capacity planning, surge assessment, labor utilization, production re-sequencing
CESA Plant Manager	Spanish operations	Full Spanish-language operations, local reporting, production management

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6. AI Agent Architecture

6.1 Agent Identity

Property	Value
Name	Atlas AI
Symbol	✦ (sparkle star)
Framework	Microsoft Agent Framework & Microsoft Foundry
Models	GPT-5.4-mini (orchestration, routing), KIMI2.6 (complex reasoning)
Protocol	SSE (Server-Sent Events) for real-time streaming
Personality	Data-driven, executive-concise, action-oriented
Language	Responds in user's selected language (en/fr/es)

6.2 Specialist Agents

Agent	Responsibility	Key Outputs
Demand Sensing	Program rates, OEM schedule changes, MRO forecasting	SKU-level 8-week forecasts with confidence bands
Inventory Risk	Stock positions, DOS, risk scoring, cert expiry	Risk matrix, predictive stockout alerts
Supply Constraint	Supplier lead times, capacity, material allocation	Reliability scores, delayed PO flags

Replenishment	Purchase orders, safety stock, production priorities	Prioritized action cards with KPI impact

6.3 Scenario Pipeline (5-Stage)

Stage	Agent	Output
1	Scenario Planner	Classify and structure the scenario
2	Impact Analyzer	Quantitative modeling (demand, inventory, supply, production)
3	Mitigation Designer	Generate alternatives (suppliers, production surge)
4	Risk Assessor	Probability and severity assessment
5	Synthesizer	Executive summary with decision points

6.4 Report Generation Pipeline (5-Agent)

Stage	Agent	Output
1	Planner	Outline slides/sections with allocation
2	Content Writer	Fill with real data and narrative
3	Designer	Apply HD branding, layout validation
4	Critic	Score quality 1-10, identify issues
5	Repair	Fix issues if score < 7 (max 2 iterations)

6.5 Query Routing

Query Type	Route
Demand / forecast / program rate	Demand Sensing Agent
Stock / inventory / cert expiry	Inventory Risk Agent
Supplier / capacity / material	Supply Constraint Agent
Replenishment / orders	Replenishment Agent
Contract / PO pricing	Contract Validation Tool
Labor / efficiency / overtime	Labor Utilization Dashboard
What-if / scenario	5-Stage Scenario Pipeline
Generate report / deck	5-Agent Report Pipeline

7. Trilingual Internationalization

The platform supports full trilingual operation across all UI elements:

Element Type	Coverage
Navigation & page titles	All 10 modules
Table column headers	Every data table across all pages
Chart legends & labels	All Recharts visualizations
KPI card titles	Dashboard, Labor, all modules
Badge text	Critical/Warning/Safe (translated)
Scenario results	All 4 tabs, history, comparison
Dialog text & buttons	Confirmations, approve, cancel
Atlas AI responses	Language-aware LLM prompting
Report content	Generated in selected language
Chat welcome & chips	Page-specific per language

Implementation: 300+ translation keys via custom React i18n context.
Language switch: One-click rotation (EN → FR → ES → EN) in sidebar.

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8. Key Differentiators

Feature	Description
AI drives, not decorates	Every page is an Atlas AI workflow — not a chatbot sidebar bolted on
Multi-agent orchestration	4 specialists + orchestrator with parallel execution
5-stage scenario pipeline	Quantitative what-if replacing days of manual modeling
Trilingual operation	Complete EN/FR/ES coverage — UI, charts, tables, AI responses
Context-aware chat	Welcome messages and chips change per page and language
Decision-ready output	Recommendations with costs, timelines, and KPI impact — not just data
One-click reports	7 templates across 4 formats, branded, generated in seconds
Microsoft Foundry	Enterprise-grade agent framework with GPT-5.4-mini + KIMI2.6 models
Real AI output	All agent responses are LLM-generated — not templates or rules

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9. Technology Stack

Layer	Technology
Frontend	React 19, TypeScript, Vite, Fluent UI 9

Backend	Python FastAPI, Microsoft Agent Framework
AI Platform	Microsoft Foundry
AI Models	GPT-5.4-mini (orchestration), KIMI2.6 (complex reasoning)
Streaming	SSE (Server-Sent Events) for real-time agent communication
Charts	Recharts (area, line, bar, pie)
Document Gen	PptxGenJS, docx, jsPDF, ExcelJS
i18n	Custom React context (300+ keys, 3 languages)
Session	UUID-based multi-user isolation
Deployment	Vite dev server (5174) + FastAPI (8001)

10. Demonstrated Value (PoC Metrics)

Based on synthetic data simulating Heroux-Devtek's operational environment:

Metric	Value
Scenario analysis time	2-3 days → 30 seconds (99% reduction)
S&OP deck preparation	6+ hours → 45 seconds (99% reduction)
Morning supply review	45 minutes → 2 minutes (95% reduction)
Stockout detection	Reactive → 8-week predictive horizon
Contract compliance	Monthly spot-check → real-time continuous
Labor reporting	Manual end-of-week → daily automated (80% reduction)
Report formatting effort	40% of planner time → zero (100% elimination)
Cross-site communication	English-only → native language (zero translation overhead)

11. Roadmap

Phase	Scope	Status
Phase 1	Full prototype — mock data, all agent pipelines, 10 modules, trilingual UI	**PoC Complete**
Phase 2	SAP/ERP data integration, live inventory feeds, Windchill PLM link	Q3 2026
Phase 3	MES integration (production actuals), LIMS quality data	Q4 2026
Phase 4	Automated PO creation (agent-initiated procurement)	Q1 2027
Phase 5	Predictive maintenance integration (IoT sensor feeds)	Q2 2027

12. Summary

ATLAS transforms Heroux-Devtek's supply chain planning from a fragmented, reactive, spreadsheet-driven process into a unified, AI-powered decision engine. By combining multi-agent orchestration with domain-specific aerospace knowledge (program rate impacts, titanium supply dynamics, NADCAP certification requirements, multi-facility production), the platform delivers measurable value across all 5 operational personas — from S&OP planners managing daily inventory decisions to leadership needing board-ready scenario analysis.

Key capabilities in v1.1:

- **Microsoft Agent Framework orchestration** — 4 specialists + orchestrator powered by GPT-5.4-mini and KIMI2.6
- **5-stage scenario pipeline** — quantitative what-if replacing 2-3 days of manual modeling
- **Trilingual operation** — complete EN/FR/ES coverage for global team collaboration
- **7 report templates** across PPTX/DOCX/PDF/XLSX with HD branding and zero formatting
- **Context-aware Atlas AI** — page-specific, language-aware conversational interface
- **300+ translated elements** — tables, charts, badges, dialogs, settings all adapt to language

The platform is designed to complement and augment existing ERP/MES systems rather than replace them, making it deployable alongside current workflows with minimal disruption.

Prepared by Atlas AI Platform Team | Heroux-Devtek Inc. | June 2026